Group: Royal Master

Case 2 Jamming

1. Abstract

This report shows the implementation and analysis of the directional jamming in a drone simulation environment. The main target is to see how concentrated jamming affects drone communication with the ground control station (GCS). Performance metrics like latency, packet loss, signal-to-noise ratio (SNR), and throughput are tested to evaluate the effectiveness of this attack

2. Introduction

Jamming attacks have a significant risk to cyber-physical systems (CPS), for example, the drones, through interfering with their communication channels. In the report of case 2, we executed and examined four types of distinct jamming attack methods. They are: Continuous Wave (CW) Jamming, Sweeping Jamming, Pulsed Noise Jamming, and Directional Jamming. The target and the aim was to examine how these types of attacks act on drone navigation and communication through the ADS-B (Mode S Extended Squitter transponder - 1090ES) channel.

2. Jamming Attacks Types:

As we stated previously we have four types of attack and they are:

1. ***Continuous Wave (CW) Jamming:*** Sends a continuous noise signal that interrupts the communication.
2. ***Sweeping Jamming:*** Utilizes a shifting frequency to result in broken signal loss.
3. ***Pulsed Noise Jamming:*** Sends small bursts of interference, disrupting messages.
4. ***Directional Jamming:*** Choose a particular area, stopping signals within its range.

3. Jamming Implementation

The directional jamming was presented to selectively damage communication between a jammer and a aimed drone with not affecting the whole network. This is not the same as the usual broad-range jamming; this method concentrates on interference within a particular angle range. So to integrate jamming mechanisms into the Drone-Sim project, our group changed several simulation scripts and the Jammer class. Every single script holds a different jamming method and changes the UAV communication accordingly:

- **(Jammer.py)**: Handels the possibility of jamming, noise intensity, and signal degradation parameters.
- **(Cw jamming)**: Simulates a constant increase in jamming strength through the time.
- **(Sweeping jamming)**: Integrates a regular difference in interference, mimicking frequency-hopping disruptions.
- (**Pulsated noise jamming**): Produce unknown bursts of interference to interrupt and damage the communication intermittently.
- (**Directional jamming**): Limit signal disruption to a known geographic area, affecting the drones only within the range.

4. Results & Analysis:

To assess the impact of several types of jamming techniques, we examine four key performance metrics:

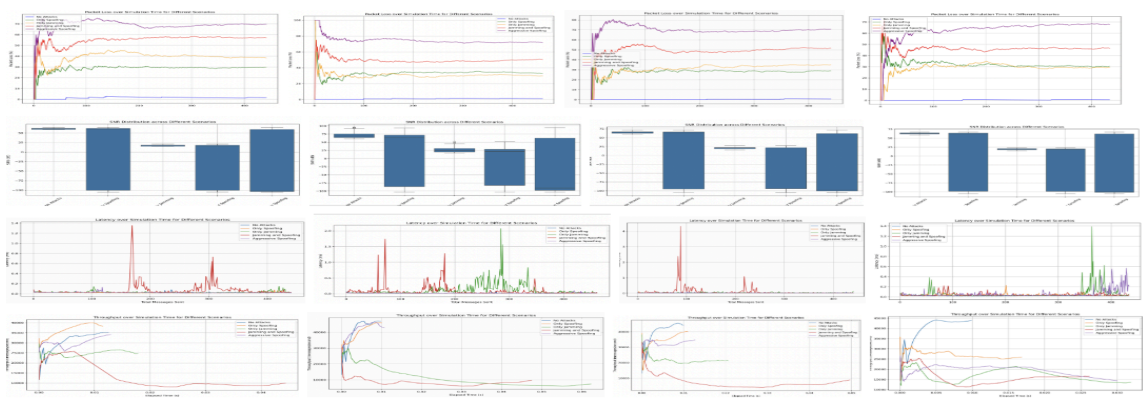
Latency: Calculate the delay in communication.

Packet Loss: Shows the percentage of lost data packets.

Signal-to-Noise Ratio (SNR): Represents signal strength opposed to interference.

Throughput: Calculate the rate of effective message transmission.

Metric	CW Jamming	Sweeping Jamming	Pulsed Noise Jamming	Directional Jamming
Latency	Gradual enhance	Periodic spikes	Small bursts	Little delay in the jammed space
Packet Loss	30 to 40%	30-35%	20-30%	25-32%
SNR Impact	Constant reduce	Fluctuates	Inconsistent variations	Low signal in the area that is jammed
Throughput	Reducing over time	Falls while interference	Small fluctuations	Reduced in the jammed area and it is normal anywhere else.



5. ***Jamming Impact***

Jamming Type	Cyber Impact	Physical Impact
CW Jamming	Constant interference, big packet loss, degraded SNR.	Losing the control, undependable navigation, potential fail-safe mode activation.
Sweeping Jamming	Cycle signal loss, unpredictable interference.	Flight way variation, late control response.
Pulsed Noise Jamming	Small bursts of interference, irregular data loss.	Fast navigation failures, late UAV responses.
Directional Jamming	Make the movement of the signals hard in one area, medium packet loss.	Impacts on the drone only in that specific area, there is no effect outside of the area.